Visual_Speech_Recognition_for_Multiple_Languages Public

About

Visual Speech Recognition for Multiple Languages

#visual-speech-recognition #lipreading

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Python 100.0%

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Pingchuan Ma update readme · 5e1405d · 3 years ago · 14 Commits

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infer.py	add AutoAVSR models and refactor the c...	3 years ago
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Visual Speech Recognition for Multiple Languages

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Authors

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Update

README License

`2023-06-16` : We have released our training recipe for AutoAVSR, see here.

`2023-03-27` : We have released our AutoAVSR models for LRS3, see here.

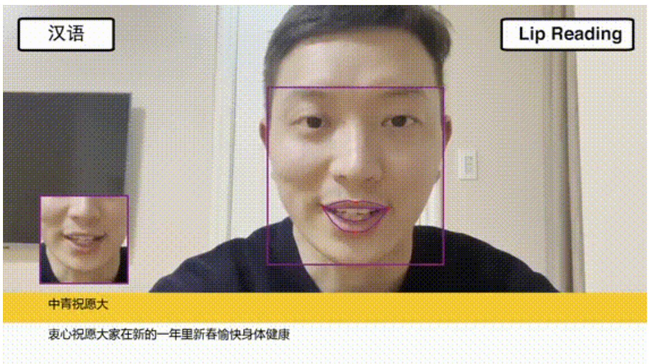
Introduction

This is the repository of Visual Speech Recognition for Multiple Languages, which is the successor of End-to-End Audio-Visual Speech Recognition with Conformers. By using this repository, you can achieve the performance of 19.1%, 1.0% and 0.9% WER for automatic, visual, and audio-visual speech recognition (ASR, VSR, and AV-ASR) on LRS3.

Tutorial

We provide a tutorial Open in Colab to show how to use our Auto-AVSR models to perform speech recognition (ASR, VSR, and AV-ASR), crop mouth ROIs or extract visual speech features.

Demo

English -> Mandarin -> Spanish	French -> Portuguese -> Italian
	

Youtube | Bilibili

Preparation

1. Clone the repository and enter it locally:

```
git clone https://github.com/mpc001/Visual_Speech_Recognition_for_Multiple_Languages
cd Visual_Speech_Recognition_for_Multiple_Languages
```

2. Setup the environment.

```
conda create -y -n autoavsr python=3.8
conda activate autoavsr
```

3. Install pytorch, torchvision, and torchaudio by following instructions here, and install all packages:

```
pip install -r requirements.txt
conda install -c conda-forge ffmpeg
```

4. Download and extract a pre-trained model and/or language model from model zoo to:

- `./benchmarks/${dataset}/models`
- `./benchmarks/${dataset}/language_models`

5. [For VSR and AV-ASR] Install RetinaFace or MediaPipe tracker.

Benchmark evaluation

```
python eval.py config_filename=[config_filename] \
               labels_filename=[labels_filename] \
               data_dir=[data_dir] \
               landmarks_dir=[landmarks_dir]
```

- `[config_filename]` is the model configuration path, located in `./configs` .
- `[labels_filename]` is the labels path, located in `${lipreading_root}/benchmarks/${dataset}/labels` .
- `[data_dir]` and `[landmarks_dir]` are the directories for original dataset and corresponding landmarks.
- `gpu_idx=-1` can be added to switch from `cuda:0` to `cpu` .

Speech prediction

```
python infer.py config_filename=[config_filename] data_filename=[data_filename]
```

- `data_filename` is the path to the audio/video file.
- `detector=mediapipe` can be added to switch from RetinaFace to MediaPipe tracker.

Mouth ROIs cropping

```
python crop_mouth.py data_filename=[data_filename] dst_filename=[dst_filename]
```

- `dst_filename` is the path where the cropped mouth will be saved.

Model zoo

Overview

We support a number of datasets for speech recognition:

- [x] Lip Reading Sentences 2 (LRS2)
- [x] Lip Reading Sentences 3 (LRS3)
- [x] Chinese Mandarin Lip Reading (CMLR)
- [x] CMU Multimodal Opinion Sentiment, Emotions and Attributes (CMU-MOSEAS)
- [x] GRID
- [x] Lombard GRID
- [x] TCD-TIMIT

AutoAVSR models

▼ Lip Reading Sentences 3 (LRS3)

Components	WER	url	size (MB)
Visual-only			
-	19.1	GoogleDrive or BaiduDrive(key: dqsy)	891
Audio-only			
-	1.0	GoogleDrive or BaiduDrive(key: dvf2)	860
Audio-visual			
-	0.9	GoogleDrive or BaiduDrive(key: sai5)	1540
Language models			
-	-	GoogleDrive or BaiduDrive(key: t9ep)	191
Landmarks			
-	-	GoogleDrive or BaiduDrive(key: mi3c)	18577

VSR for multiple languages models

▼ Lip Reading Sentences 2 (LRS2)

Components	WER	url	size (MB)
Visual-only			
-	26.1	GoogleDrive or BaiduDrive(key: 48l1)	186
Language models			
-	-	GoogleDrive or BaiduDrive(key: 59u2)	180
Landmarks			
-	-	GoogleDrive or BaiduDrive(key: 53rc)	9358

▼ Lip Reading Sentences 3 (LRS3)

Components	WER	url	size (MB)
Visual-only			
-	32.3	GoogleDrive or BaiduDrive(key: 1b1s)	186
Language models			
-	-	GoogleDrive or BaiduDrive(key: 59u2)	180
Landmarks			
-	-	GoogleDrive or BaiduDrive(key: mi3c)	18577

▼ Chinese Mandarin Lip Reading (CMLR)

Components	CER	url	size (MB)
Visual-only			
-	8.0	GoogleDrive or BaiduDrive(key: 7eq1)	195
Language models			
-	-	GoogleDrive or BaiduDrive(key: k8iv)	187
Landmarks			
-	-	GoogleDrive or BaiduDrive(key: 1ret)	3721

▼ CMU Multimodal Opinion Sentiment, Emotions and Attributes (CMU-MOSEAS)

Components	WER	url	size (MB)
Visual-only			
Spanish	44.5	GoogleDrive or BaiduDrive(key: m35h)	186
Portuguese	51.4	GoogleDrive or BaiduDrive(key: wk2h)	186
French	58.6	GoogleDrive or BaiduDrive(key: t1hf)	186
Language models			
Spanish	-	GoogleDrive or BaiduDrive(key: 0mii)	180
Portuguese	-	GoogleDrive or BaiduDrive(key: l6ag)	179
French	-	GoogleDrive or BaiduDrive(key: 6tan)	179
Landmarks			
-	-	GoogleDrive or BaiduDrive(key: vsic)	3040

▼ GRID

Components	WER	url	size (MB)
Visual-only			
Overlapped	1.2	GoogleDrive or BaiduDrive(key: d8d2)	186
Unseen	4.8	GoogleDrive or BaiduDrive(key: ttsh)	186
Landmarks			
-	-	GoogleDrive or BaiduDrive(key: 16l9)	1141

You can include `data_ext=.mpg` in your command line to match the video file extension in the GRID dataset.

▼ Lombard GRID

Components	WER	url	size (MB)
Visual-only			
Unseen (Front Plain)	4.9	GoogleDrive or BaiduDrive(key: 38ds)	186
Unseen (Side Plain)	8.0	GoogleDrive or BaiduDrive(key: k6m0)	186
Landmarks			
-	-	GoogleDrive or BaiduDrive(key: cusv)	309

You can include `data_ext=.mov` in your command line to match the video file extension in the Lombard GRID dataset.

▼ TCD-TIMIT

Components	WER	url	size (MB)
Visual-only			
Overlapped	16.9	GoogleDrive or BaiduDrive(key: jh65)	186
Unseen	21.8	GoogleDrive or BaiduDrive(key: n2gr)	186
Language models			
-	-	GoogleDrive or BaiduDrive(key: 59u2)	180
Landmarks			
-	-	GoogleDrive or BaiduDrive(key: bnm8)	930

Citation

If you use the AutoAVSR models training code, please consider citing the following paper:

```
@inproceedings{ma2023auto,
  author={Ma, Pingchuan and Haliassos, Alexandros and Fernandez-Lopez, Adriana and Chen, Honglie and Petridis, St
  booktitle={IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)},
  title={Auto-AVSR: Audio-Visual Speech Recognition with Automatic Labels},
  year={2023},
}
```

If you use the VSR models for multiple languages please consider citing the following paper:

```
@article{ma2022visual,
  title={{Visual Speech Recognition for Multiple Languages in the Wild}},
  author={Ma, Pingchuan and Petridis, Stavros and Pantic, Maja},
  journal={{Nature Machine Intelligence}},
  volume={4},
  pages={930--939},
  year={2022}
  url={https://doi.org/10.1038/s42256-022-00550-z},
  doi={10.1038/s42256-022-00550-z}
}
```

License

It is noted that the code can only be used for comparative or benchmarking purposes. Users can only use code supplied under a License f
non-commercial purposes.

Contact

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```